

Fix a field $\mathbb{F}$ (e.g. $\mathbb{F} = \mathbb{C}$ or $\mathbb{R}$) for $0 \leq k \leq n$, and recall the definition of the grassmannian

$$Gr_{k,n} = \left\{ \begin{array}{l} k \times n \text{ matrices} \\ \text{of rank } k \end{array} \right\} / \{ \text{row operations} \}$$

$$Gr_{k,n} = \left\{ \left[\begin{array}{ccc|c} 1 & & & 0 \\ & \ddots & & \\ & & 1 & \\ 0 & & & \end{array} \right] \right\} \cup \{ \text{something else} \}$$

$$\overset{\sim}{\mathbb{F}^{k(n-k)}} \leftarrow \text{"affine space of dim. } k(n-k) \text{"}$$

let's review Gaussian elimination...

$$A \xrightarrow{\text{row ops}} \text{RREF (reduced row echelon form)}$$

EX: Committed from written notes, included in typed)

↳ in class: 5×11 matrix

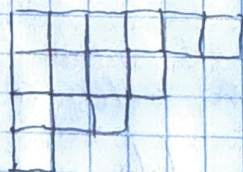
* each matrix A has a unique RREF.

* number of pivots (columns w/ a 1 entry) should match the rank

If we remove pivot columns, we then get a shape of "stars" which let us define a corresponding young diagram

$$\begin{bmatrix} 1 & x & x & x & x & x & x \\ 0 & 0 & 1 & x & x & x & x \\ 0 & 0 & 0 & 1 & x & x & x \\ 0 & 0 & 0 & 0 & 1 & x & \end{bmatrix}$$

remove pivots
and change *
to boxes



$$\rightarrow \lambda = (6, 4, 3, 1)$$

where our young diagram corresponds to a partition $\lambda \in k \times (n-k)$

Def'n (Schubert Cells):

$$\rightarrow X_\lambda^0 = \{ \text{points in } Gr_{k,n} \text{ whose RREF has pivots in position } I(\lambda) = \lambda \}$$

Def'n (Schubert Variety):

$$\rightarrow X_\lambda = X_\lambda^0$$

Thm: $Gr_{kn} = \bigsqcup_{\lambda \in k \times (n-k)} \chi_{\lambda} \chi_{\lambda}^0$ and $\chi_{\lambda}^0 \cong \mathbb{F}^{|\lambda|}$ ← # of boxes in young diagram

for $\lambda, \mu \in k \times (n-k)$, $\lambda \leq \mu$ iff $\lambda \subseteq \mu$.

This process is called Schubert decomposition.

Corollary:

1) for $\mathbb{F}_q = \mathbb{F}_q$ we have $Gr_{kn}(\mathbb{F}_q) = \sum_{\lambda \in k \times (n-k)} q^{|\lambda|} = \binom{n}{k}_q$

↳ q -binomial coefficients, one of the many " q -analogs" combinatorialists care about.

2) $\chi_{\lambda} \subseteq \chi_{\mu}$ iff $\lambda \subseteq \mu$.

* when $\mathbb{F} = \mathbb{C}$, we can calculate the cohomology $H^*(Gr_{kn}(\mathbb{C}))$,

↳ recall the key example for $Gr_{12} = \mathbb{P} = \mathbb{R} \cup \{\text{point}\}$

The more common definition for a Schubert cells is given in terms of the complete flag of subspaces in a field $\mathbb{F}^n$.

$$V_0 = \{0\} \subset V_1 \subset V_2 \subset \dots \subset V_n \cong \mathbb{F}^n$$

where V_i is defined by the linear span of standard vectors $e_1, e_2, \dots, e_{n-i}$ (last i coordinate vectors)

Let V be a k -dimensional vector space in $\mathbb{F}^n$ (row space of A)

Lemma: fixing pivot positions $\{i_1, \dots, i_k\}$ of RREF is equiv to fixing the vectors d

$$(d_1, \dots, d_n) \text{ such that } d_i = \dim(V \cap V_i)$$

with $0 \leq d_1 \leq \dots \leq d_n = k$ and $d_{i+1} \leq d_i + 1$ for all i .

Def'n (Schubert cells - part 2):

↳ $\chi_{\lambda}^0 = \{V \in Gr_{kn} \mid \dim(V_i \cap V) = d_i \text{ for } i = 1, \dots, n\}$

$$\chi_{\lambda} = \{V \mid \dim(V_i \cap V) \geq d_i \text{ for } i = 1, \dots, n\}$$

exercise: carefully check the second condition - strict ineq. in exceptional cases

Def'n (Plücker coordinates on $Gr_{k,n}$):

↳ consider maximal minors of a matrix A

$$\Delta_I = \det \begin{pmatrix} k \times k \text{ submatrix} \\ \text{in cols. } i_1, i_2, \dots, i_k \end{pmatrix}$$

where $I = \{i_1, \dots, i_k\} \in \binom{[n]}{k}$ where $[n] = \{1, \dots, n\}$
and $\binom{[n]}{k}$ is all k -elt subsets of $[n]$,

Lemma: for all $I \in \binom{[n]}{k}$ and $\lambda = \lambda(I)$, we have

$$X_\lambda^0 = \left\{ \text{points of } Gr_{k,n} \text{ s.t.} \right. \\ \left. I = \text{lex. minimal } k\text{-tuple s.t. } \Delta_I \neq 0 \right\}$$

→ we can figure out pivots of A without Gaussian elimination
by computing determinants of maximal minors

↳ this is not more efficient, but has other
things to offer.

Matroidal (or Gelfand-Serganova) stratification of $Gr_{k,n}$

→ for $M \in \binom{[n]}{k}$ nonempty, we have

$$\text{"strata"} \rightarrow S_M = \left\{ \text{points of } Gr_{k,n} \text{ s.t.} \right. \\ \left. \Delta_I \neq 0 \text{ iff } I \in M \right\}$$

$$\text{then } Gr_{k,n} = \bigsqcup S_M$$

this view helps us get at many important bits of combinatorial data:

↳ dimensions, intersections, etc.

→ ordering matters (for coordinates)

Now, we can say S_n acts on $Gr_{k,n}$ by permutations of columns of A

Thm: Matroid strata S_M are common refinements of all permuted
Schubert cells

$$\omega(X_\lambda^0) \text{ for } w \in S_n.$$